GLA University, Mathura

Second Mid Term Examination, 2013-2014

Course: - B.Tech | **II-Year, III-Sem.**
Subject: -Mathematics-III (AHM -201) | **Uni. Roll No:-**
Time: - 1 Hr. 30 Minutes | **Total Marks: - 20**

Notes:-

1) Attempt **ALL** groups.
2) Attempt **ANY FOUR** questions from Group A, **ANY THREE** questions from Group B and **ANY TWO** questions from Group C.
3) All questions of a group should be answered at one place.
4) Answer should be brief and to-the-point.
5) Any missing or wrong data may be assumed suitably giving proper justification.
6) Figures on the right-hand side margin indicate full marks.

Section –A (1x4 = 4 marks)

Attempt any four questions:

Q.1 If $F\{e^{-x^2}\} = \sqrt{\pi}e^{-p^2/4}$ then find Fourier transform of $e^{-4(x-3)^2}$

Q.2 If Fourier cosine transform of $\frac{1}{1+x^2}$ is $\frac{\pi}{2}e^{-p}$, then find the value of $\int_0^\infty \frac{x\sin px}{1+x^2}dx$

Q.3 If $F(x) = x^3$, then prove that $x^3 = \frac{2}{\pi}\int_0^\infty\int_0^\infty t^3 \sin\lambda t \sin\lambda x\, dt\, d\lambda$

Q.4 Find the Z- transform of sequence $f(k) = \begin{cases} 5^k, & k<0 \\ 2^k, & k\geq 0 \end{cases}$.

Q.5 If Z{f(k)}=F(z),then find $Z\{|a^k f(k)|\}$

Section –B (2x3 = 6 marks)

Attempt any three questions

Q.1 Find F(x), if its Fourier sine transform is $\begin{cases} \sin p, & 0<p<\pi \\ 0, & p\geq\pi \end{cases}$

Q.2 Find the Fourier cosine transform of the function $\frac{e^{ax}+e^{-ax}}{e^{\pi x}+e^{-\pi x}}$

Q.3 Using Fourier sine integral, show that $\int_0^\infty \left(\frac{1-\cos\lambda\pi}{\lambda}\right)\sin\lambda x d\lambda = \begin{cases} \frac{\pi}{2}, & 0<x<\pi \\ 0, & x>\pi \end{cases}$

Q.4 Find the inverse Z-tranform of $F(z) = \frac{4z}{(z-a)}$ for (i) $|Z|>|a|$ (ii) $|Z|<|a|$

Section –C (5x2 = 10 marks)

Attempt any two questions

Q.1 Find Fourier sine transform of e^{-ax}/x and then find Fourier sine transform of $1/x$.

Q.2 Determine the distribution of temperature in the semi infinite medium $x \geq 0$ .when the end $x = 0$ is maintained at zero temperature and the initial distribution of temperature is F(x)

Q.3 Using Fourier integral formula, prove that

$$\int_0^{\infty}\left(\frac{\lambda^2+2}{\lambda^4+4}\right)\cos\lambda x\,d\lambda = \frac{\pi}{2}e^{-x}\cos x, \quad x>0$$

Q.4 Using the Z-transform, solve the difference equation

$$y_k - \left(\frac{1}{3}\right)y_{k-1} = \left\{\left(\frac{1}{3}\right)^k\right\} \quad k \geq 0,\ y(0)=0$$

[redacted]

GLA University
Mid Term Exam
B. Tech: II Year, 3rd semester [CSE, EC, EE & EN]
Ethics & Values (AHE: 201)

Time: 1.30 Hours **Marks: 20**

This paper consists of three passages. Carefully read the passages and answer the following multiple choices questions.
Full marks (assigned) will be given for a complete answer that means all correct choices.
Half mark (assigned) will be given for an incomplete answer that means one or more correct choices but not the all.
Zero (0) will be given if one clubs an incorrect choice with one or more correct choices.
Marks are assigned before passages.

Passage-2 (1 X10 = 10 Marks)

Sometimes what seems to be absolutely immoral becomes the only way to be moral. There comes a clash between the morality and the necessity. And, it raises a question: Can we exonerate someone from the guilty if someone has acted out of necessity? Queen Vs Dudley and Stephens is one of the most debated cases in the law schools, which had established an example, in the British common law, that necessity is not a defense for a charge of murder. The case is also popular as The Life Boat Case (1884). The English yacht *Mignonette* was to be transported to Australia with crew of four: Thomas Dudley, the captain; Edwin Stephens; Edmund Brooks; and Richard Parker, the cabin boy. Case is very much clear to you that Dudley and Stephenson went for trail, while Brooke turned to a witness. Now, judge has to decide whether the killing and cannibalism done by the three on life-boat for saving their lives was justified. Stephenson argued that they have done wrong but it was the only way out and was an unavoidable necessary action. He reasoned that necessity was to save the three lives so he killed Parker. Judge: why did you kill parker? Dudley: Parker was orphan and was ill. We other members have family. We thought if Parker would die nobody would have sorrow but any one of us would die then his family would be in a great pain. One had to die in this scenario and with this necessity parker was the right candidate to be killed. Judge came with the point that after four days you guys got rescued and you could have survived for four days without killing Parker then in this possibility killing Parker neither proved necessity not the utility. And, if you had no idea when you will get rescue then killing Parker does not make any sense as there is no guarantee that after subsisting on Parker you will not have to kill another. Killing Richard Parker makes a sense of necessity when there is certainty that the rescue will come and for that time you cannot survive without killing Parker. It was the argument from necessity that makes clear that necessity cannot be sought in the view of uncertainty. Stephenson got disproved on the ground of necessity moved to another argument called argument from intention that we did not want to kill Richard Parker. We wanted to save three lives. Judge asked a question to Stephenson; was it you will-full action? Stephenson replied no then judge said why did you kill then Parker? You could have killed anyone or could have sacrificed your own life for saving the other three. Stephenson was willing the action as he minded first his interest in saving his own life. The arguments presented by the Dudley and Stephenson based on maximum life saving, survival, sacrifice and life defense did not suffice them to excuse from the charges of murder and cannibalism. The jury decided the sentence to death for both Dudley & Stephenson but that later changed by the interference of the crown into imprisonment. However the case offers a thrilling moral debate

on the point whether necessity outweigh the morality. But in this case it was disproved. And, proved a murder is murder and it cannot be right even looked by the glaring glasses of morality. However, the necessity disapproved in the case can be further debated if we add some facts like moral agency, consent and the sacrifice. Once these facts are added in the case it seems that it improves the case of necessity by upgrading the moral degrees to a certain extent. However, one thing is obvious that killing Richard Parker in such a manner Dudley and Stephenson did not respect the sense of morality as they did not respect the agency of Parker and killed him against his free will. Parker was not consented at all and it was not his choice. Indeed, if we add consent of Parker it includes in some ways the agency of Parker also. But, consent does not mean that they should have been convinced are informed Richard Parker in advance, that they are going to kill him. It is not only the consent in this case that matters but also the due process. The right process to take the consent of Richard parker could be the lottery option that Stephenson once proposed to all. Nevertheless, drawing lottery would be all right if they all would have agreed on it. The agreement could have included all members will. And, then the member whose name would have drawn out in favor of death would be considered as a matter of sacrifice to save maximum life. Thus, the sacrifice differs from murder as in sacrifice one gives one's life by one's own will for saving others and in murder one's life is taken by the other against one's will. Hence, adding lottery option to this case takes it to the proximity to the necessity and the death of any member to save other's lives proves sacrifice. But, some author would question such necessity even if it includes agency and consent: Can we draw lottery by putting someone's life in question? They are the theorists who are categorical in their views on the question of rights and wrongs. They strongly maintain that the value of life is not comparable and cannot be sacrificed on maximum numbers. They advocate deontological view on morality. Morality for them is not the matter of just utility. They argue that life is valuable in itself. It does not increase or decrease by the numbers of people associated with it or not. For them Parker's life was equally important and not less than any ways even if you compare with the three or you add the lives of their family members too. For such theorists morality does not depend on the consequences. They locate the morality in the nature of an action. The moral worth of an action according to them lies in the very nature of an action. Hence, they would condemn the case of cannibalism as done out of necessity and would say there cannot be any such necessity beyond the morality. For them the case shows implicit immorality as it compares the value of life with external factors. Necessity means only in a moral framework and it means it has to be moral necessity and not the ordinary necessity.

Questions:

1) How did argument of necessity disprove in this case?

- Because Dudley and Stephenson were sure about rescue.
- Dudley and Stephenson had no idea about the rescue.
- Stephenson killed Parker in the view of un-surety.
- Necessity cannot be proved in the view of uncertainty.

2) Which of the following statements are true about the case?

- Brooke was witness in this case.
- Dudley murdered Richard Parker.
- It was the case of consented cannibalism
- This case is also known as the Life-Boat case.

3) The necessity can be proved in this case with the addition of the following facts:

- Moral Agency
- Sacrifice
- Consent
- None of these.

4) What does consent include?

- One's free will
- One's agency
- One's interest.
- None of these.

5) What did Stephenson try to prove by argument from intention?

- He did not want to kill Richard Parker.
- He wanted to save three lives.
- He wanted to save himself.
- All above.

6) Which of the following arguments were not offered by Stephenson in his defense?
- The argument from maximum lives saving.
- The argument from uncertainty.
- The argument from necessity.
- The argument from sacrifice.

7) Why did the lottery option might be the right procedure for seeking consent?
- Because it would have been followed by the agreement of everyone.
- It would have involved everyone's agency.
- It would have proved Parker's killing right.
- None of these.

8) Necessity cannot go beyond the morality. What does this categorical remark mean to this case?
- Necessity exists within the morality.
- Value of life cannot be compared with external factors.
- It is a moral necessity.
- Necessity may go beyond morality.

9) Which of the following statements are true about categorical view to this case?
- Value of life is not comparable.
- Life cannot be sacrificed for maximum numbers.
- Morality is not a matter of utility.
- Life in valuable in itself.

10) For some theorists even if the necessity includes the consent and agency would not be alright as life issues cannot be decided by lottery. What can we infer about these theorists?
- They advocate deontological view of morality.
- For them Parker's life was equally important as the life of others.
- They locate morality in the nature of an action.
- They condemn the view of necessity that goes beyond the morality.

Passage:2

Bentham, an English moral philosopher and legal reformer, founded the doctrine of utilitarianism. Its main idea is simply stated and intuitively appealing: The highest principle of morality is to maximize happiness, the overall balance of pleasure over pain. According to Bentham, the right thing to do is whatever will maximize utility. By "utility", he means whatever produces pleasure or happiness, and whatever prevents pain or suffering. For Bentham, pleasure is pleasure and pain is pain. The only basis for judging one experience better or worse than another is the intensity and duration of the pleasure or pain it produces. Bentham recognizes no qualitative distinction among pleasures. "The quantity of pleasure is equal", he writes, "push-pin is as good as poetry." The refusal to distinguish higher from lower pleasure is connected to Bentham's belief that all values can be measured and compared on a single scale. But some object to utilitarianism on precisely this point: they believe that some pleasures really are "higher" than others. If some pleasures are worthy and others base, they say, why should society weigh all preferences equally, much less regard the sum of such preferences as the greatest good?

Questions:

1) Bentham's utilitarian principle goes against our common human experience because:
   - Bentham recognizes no qualitative distinction among pleasures.
   - It maximizes utility.
   - It does not care about the preferences of pleasures and weigh all preferences equally.
   - None of these.
2) What is the right thing to do according to utilitarianism?
   - Whatever will maximize happiness
   - Whatever will maximize utility
   - Whatever produces happiness or reduces suffering
   - All above.
3) Intensity and duration are the bases to judge
   - Pleasures and Pains
   - Maximum utility
   - One pleasure better or worse than another
   - None of these.
4) Bentham refuses to distinguish higher from lower pleasure because
   - Pleasure is pleasure and pain is pain.
   - All values can be measured and compared on a single scale.
   - Some pleasures are higher than others.
   - All above.

Passage-3

The human faculties of perception, judgment, discriminative feeling, mental activity, and even moral preference, are exercised only in making a choice. He who does anything because it is the custom makes no choice. He gains no practice either in discerning or in desiring what is best. The mental and moral, like the muscular powers, are improved only by being used. He who lets the world, or his own portion of it, choose his plan of life for him, has no need of any other faculty than the ape-like one of imitation. He who chooses his plan for himself employs all his faculties. Mill concedes that following convention may lead a person to a satisfying life path and keep him out of harm's way. "But what will be his comparative worth as a human being?" he asks. "It really is of importance, not only what men do, but also what manner of men they are that do it. So actions and consequences are not all that matter after all. Character also counts. For J. S.Mill, individuality matters less for the pleasure it brings than for the character it reflects. "One whose desires and impulses are not his own, has no character, no more than a steam engine has character. Mill put emphasis on liberty and higher order pleasures.

Questions:

1) Conventions may guide us to the satisfying path of life but living life only by following conventions we lose;
   - Meaning of human life
   - Comparative worth as a human being
   - Self belief
   - None of these
2) Rational choice making involves:
   - Moral preference
   - Judgment
   - Discriminative feeling
   - All above
3) When author says "no more than a steam engine has character" what he means:
   - Steam engine has no character.
   - Stem engine has no desires and impulses.
   - Steam engine has character
   - None of these.
4) He who follows customs;
   - Makes moral choice
   - Makes a cultural choice
   - Makes no choice
   - Makes a choice
5) Author uses the phrase "the ape like imitation" for persons:
   - Who those follow custom.
   - Who those choose his path of life for himself.
   - Who those do not use mental and moral capacities.
   - All above
6) For J. S. Mill action and consequences are not all that matter. He put more emphasis on;
   - Individuality & Character
   - Higher Pleasure & Liberty
   - Only Individuality
   - None of these.

University Roll No. .....

GLA University, Mathura

Course – B. Tech., II-year, II-Mid Term (Odd Sem) Examination, 2013-14

Subject – Discrete Mathematics (CSE-213)

Time – 90 minutes **MM: 20**

Note:

1. Answer any four questions from Section A, any three from Section B, and any two questions from Section C.
2. All questions of a particular section should be answered collectively at one place.
3. Answer should be to-the-point and wherever required, be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate marks.

Section A (Attempt any four)

(4x1=4)

I. Determine whether the conclusion C follows from the premises H1 and H2?

H1: $\sim p$, H2: $p \leftrightarrow q \Rightarrow \sim(p \wedge q)$

II. If a class has students from three states in India, then the minimum number of students that must be chosen from the class to be sure to have two students belonging to the same state is .....

III. The number of ways in which a student can attempt a 12 question true and false examination paper if the student is free not to answer any number of questions is .....

IV. Find the type of fallacy in the argument given below:
If two sides of a triangle are equal, then the opposite angles are equal. Two sides of a triangle are not equal. Therefore the opposite angles of a triangle are not equal.

V. How many subsets with an odd number of elements does a set with ten elements have?

VI. How many committees of five with a given chairperson can be selected from 12 persons?

Section B (Attempt any three)

(3x2=6)

I. Show the following logical equivalence:

$(\sim p \wedge (\sim q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \equiv r$

II. Prove that if n is an integer and $n^3 + 5$ is odd then n is even using a proof by contradiction.

III. Obtain the PCNF of the formula given as follows:

$(\sim p \rightarrow r) \wedge (q \leftrightarrow p)$

IV. How many permutations of the letters ABCDEFG contain

a) The string CFGA
b) The strings BA and GF

Section C (Attempt any two)

(2x5=10)

I. Show that the premises "A student in this class has not read the book" and "Everyone in this class passed the first exam" imply the conclusion "Someone who passed the first exam has not read the book."

II. Translate the following using predicate logic and then write the negation of these statements:

a) There is a lawyer who never tells a lie.
b) All integers are greater than ß.

III. Prove that 5 divides n^5-n whenever n is a non-negative integer by using principle of mathematical induction.

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GLA University, Mathura

Course: -B. Tech. II-Year, II-Mid Term (Odd Sem.) Examination, 2013-14

Subject: Data Structures and Algorithms (CSE-211)

Time: - 90 Minutes **M.M:-20**

Notes:-

1. Answer **ANY FOUR** questions from Section A, **ANY THREE** from Section B and **ANY TWO** from Section C.
2. All questions of the particular section should be answered collectively at one place.
3. Answer should be to-the-point and wherever required, be supplemented with neat sketches.
4. Any missing data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate marks.

Section-A (Attempt any **four** questions) 04 x 1 = 04

1. A binary tree of depth "d" is an almost complete binary tree if_____.
2. In order to get the information stored in a Binary Search Tree in the descending order, one should traverse it in which order among Node, Left sub tree and Right sub tree?
3. One of the major drawbacks of B-Tree is the difficulty of traversing the keys in _____ order.
4. For an undirected graph with n vertices and e edges, the total sum of the degree of all vertex is equal to_____
5. The pre-order and post order traversal of a Binary Tree generates the same output. The tree can have maximum_____ nodes.
6. If a node having two children is deleted from a binary search tree, it is replaced by its_____.

Section-B (Attempt any **three** questions) 03 x 2 = 06

1. Create an extended binary tree by Huffman algorithm for the following set of weights. Also mention the codeword for each item.

Items	A	B	C	D	E	F	G
Weights	15	10	5	3	7	12	25

2. Construct a binary search tree whose nodes in postorder are given as follows:

Postorder: 1, 4, 7, 6, 3, 13, 14, 10, 8

3. Construct a B tree of order 3 for the given set of elements –

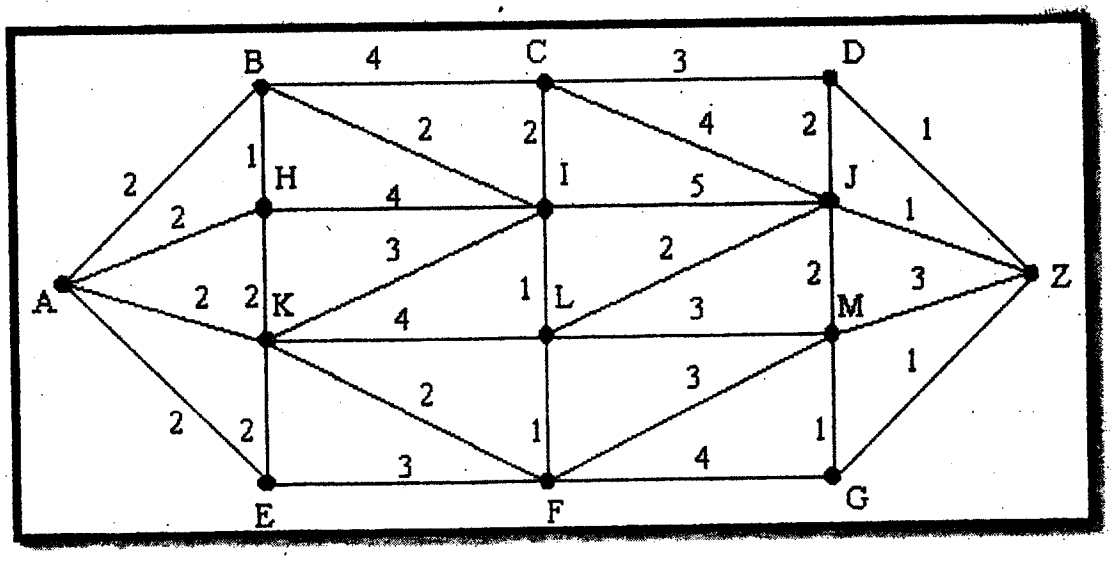
F, S, Q, K, C, L, H, T, V, W, M, R, N, P, A, B, X, Y, D, Z, E

4. Write down the Kruskal's algorithm for finding the MST of a graph.

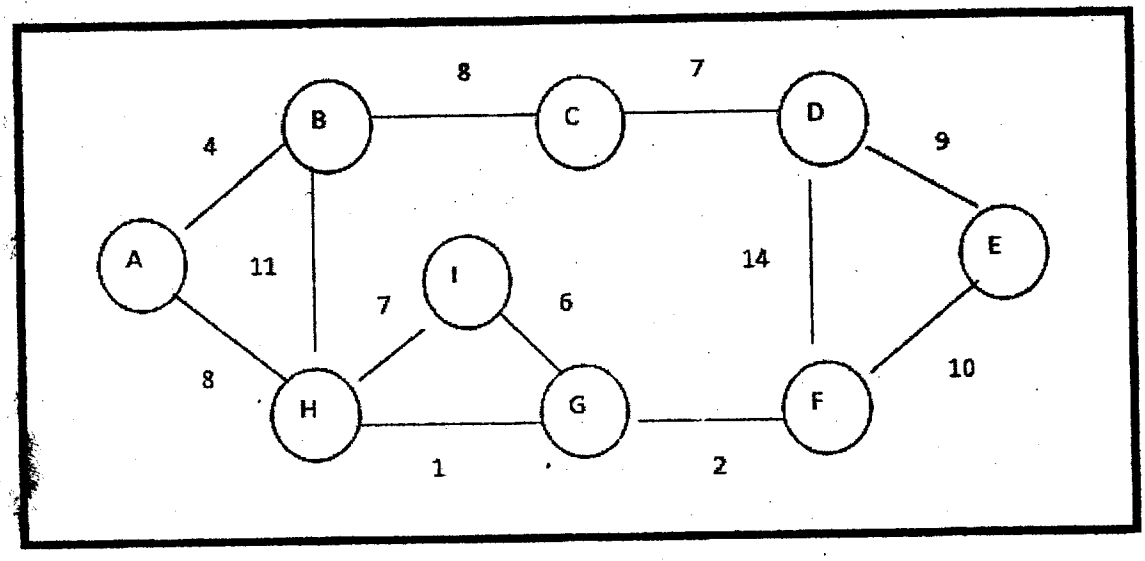
Section-C (Attempt any **two** questions) 02 x 5 = 10

1. Write down the algorithms for preorder and postorder traversals of a tree using recursion. Explain their workings with an example.

2. Find the shortest path from A to Z using Dijkstra's Algorithm.



3. Find out Minimum Spanning Tree using Prim's algorithm for the following graph.



Uni. Roll No:-.....

GLA University, Mathura

Course: -B.Tech II-Year, II-Mid Term (ODD Sem.) Examination, 2013-14

Subject: - Computer Organization (CSE-212)

Time:-90 Minutes **M.M:-20**

Notes:-

1. Answer **Any Four** questions from Group A, **Any Four** from Group B and **Any Four** from Group C.
2. All questions of the particular group should be answered collectively at one place. All Parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group-A (Answer any Four) 1x4=04

I. Perform Masking operation on: Register A 10101101 and Register B 11100110.
II. Perform 2-bit ASHL on 10110110.
III. Define sequencer.
IV. Explain Stack Frame.
V. What are the inputs to the control unit?
VI. Define type of operand.

Group-B (Answer any Three) 2x3=6

I. Starting from an initial value of R 11011101, determine the sequence of binary values in R after a logical shift-left, followed by a circular shift-right, followed by a logical shift-right and a circular shift-left.

II. The content of the top of a memory stack is 5320. The content of the stack pointer SP is 3560. A two-word call subroutine instruction is located in memory at address 1120 followed by the address field of 6720 at location 1121. What are the content of PC, SP and the top of the stack:
(a) Before the call instruction is fetched from memory?
(b) After the call instruction is executed?

III. What are the differences between hardwired and micro-programmed control unit?

IV. Explain the different pipeline hazards with suitable examples.

Group-C (Answer any Two) 5x2=10

I. Design an arithmetic circuit with one selection variable S and two n-bit data inputs A and B. The circuit generates the following four arithmetic operations in conjunction with the input carry C_{in}. Draw the logic diagram for the first two stages:

S	C_{in} 0	C_{in} 1
0	D A+B(add)	D A+1(increment)
1	D A-1(decrement)	D A+B+1(subtract)

II. Write the sequence of control steps required for the single bus structure, for each of the following instructions:
(a) Add the contents of memory location NUM to register R1.
(b) Add the contents of the memory location whose address is at memory location NUM to register R1.

III. Explain the concept of Branch Control Logic with:
(a) Two-address Field
(b) Single-address Field
(c) Variable Format
What are the applications of micro-programming.

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GLA UNIVERSITY, MATHURA

COURSE: B.TECH. II- Year, II- Mid Term (Odd Sem.), Examination, 2013-14

Sub: Object Oriented Programming (ITE 211)

Time: - *90* Minutes **M.M.:-*20***

1. All questions of the particular section should be answered collectively at one place.
2. Answer should be to-the-point and wherever required be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A (Attempt any four questions) ***4*1=4***

Q1. What is an exception? Code monitored for exceptions must be part of what statement?
Q2. Do you need to include **Java.lang** explicitly? Show how to include all of the classes in a package called **toolpack.**
Q3. What is an interface? What keyword is used to declare one? What are implements for?
Q4. How does a subclass execute its superclass's constructor? Can a call to **super** () go anywhere within a subclass's constructor?
Q5. In what two ways can you create a class that can act as a thread? What is the purpose of the **run** () metod defined by **Runnable**?
Q6. Define the use of **final** and **this** keyword in JAVA.

Section B (Attempt any two questions) ***2*3=6***

Q1. Compare and contrast: **(1+1+1)**
(a) Method Overloading and Method Overriding.
(b) Abstract class and Concrete class
(c) Class and Interface
Q2. What is java applet? Write a java program to illustrate the use of multi-catch statement. **(1+2)**
Q3. Explain the types of Layout Managers in GUI programming. **(3)**

Section C (Attempt any two questions) ***2*5=10***

Q1. What is inheritance in java? Write a program for user defined exception that checks the internal and external marks if the internal marks is greater than 20 it raise the exception" Internal marks is exceed" , if the external marks is greater than 80 exception is raised and display the message " The external marks is exceed". Create the above exception and use it in your program. **(2+3)**
Q2. Explain life cycle of a thread? Write a program to define abstract class, with two methods addition () and subtraction (). addition () is abstract method. Implement the abstract method and call that method using a program. **(2+3)**
Q3. Define package. What is the usage of javac –d option? Write the programs for creating package and importing package. **(1+1+3)**

SET-D

GLA University

Mid Term Exam

Soft Skills

Code- AHE-281

Course- B.Tech. **Year- II** **Branch-CS, EC, EN, EE, ME, CE**

Instructions :

- This Soft skills test paper consists of 90 multiple choice questions.
- The maximum marks for the test are 30.
- The total time allocated is 90 minutes.
- You have to write your answers in the Answer sheet.
- Please write your name, branch and roll number on both the Test paper and the Answer Sheet.

Passage for Questions (1- 6): Recent technological advancement in manned and unmanned undersea vehicles, overcome some of the limitations of divers equipment. Without a vehicle, divers often became sluggish and their mental concentration was limited. Because of undersea pressure that affected their mind, concentration among divers was difficult or impossible. But today, most oceanographers make observations by means of instruments that are lowered into the ocean or from samples taken from the water. Direct observations of the ocean floor are made not only by the divers, but also by deep-diving submarines. Some of these submarines can dive to depths of more than several miles and cruise at depths of 15 thousand feet. Radio equipped buoys can be operated by remote control in order to transmit information back to land-based laboratories including data about Water temperature, currents and weather. Some of mankind's most serious problems, especially those concerning energy and food may be solved with the help of observations made possible by these undersea vehicles.

1. According to the author, what are some of the problems the underwater studies eventually resolve?
a. Weather and temperature control
b. food and energy storage
c. transportation and communication problems
d. overcrowding and housing problems
e. resource shortage

2. Divers have had problems in concentrating underwater because:
a. the pressure affected their minds.
b. The vehicles they used have not been perfected.
c. they did not think clearly.
d. the pressure destroyed their mental processes.
e. of distractions while diving.

3. This passage suggests that the successful exploration of the ocean depends upon:
a. vehicles as well as divers.
b. Radio that divers use to communicate.
c. controlling currents and the weather.
d. removal of the limitations of diving equipment.
e. development of undersea vehicles.

4. With what topic is the passage primarily concerned?
a. recent technological advances.
b. communication among divers.
c. direct observation of the ocean floor.
d. undersea vehicles.
e. technological advancement of undersea vehicles.

5. How is radio-equipped buoy most likely to be operated?
a. by operators inside the vehicle and underwater.
b. by operators outside the vehicle on a ship.
c. by operators outside the vehicle on a diving platform.
d. by operators outside the vehicle in a laboratory on the shore.
e. cannot be inferred.

6. Undersea vehicles
a. are too small for a man to fit inside.
b. are very slow to respond.
c. have the same limitations that divers have.
d. make direct observations of the ocean floor.
e. are technologically primitive.

Passage for Questions (7-11): A new US study has warned that adolescents who take performance enhancing anabolic steroids are more likely to have adverse neural and behavioral consequences, like aggression and moodiness because of the steroids affect on the underdeveloped brain and the nervous system. The study, by Northeastern University in the US, centred around a brain chemical called serotonin, which is linked to mood. Lower levels of serotonin are associated with depression and aggression. For the study, experiments were carried out on a strain of Syrian hamsters. This breed has similar neurological circuitry to humans, so experts felt it might be a good model for humans in this respect. The hamsters were given a high dose of anabolic steroids over the course of a month-which corresponded to five years, repeated dosage in humans. The researchers found that the hamsters were more aggressive than those not given steroids and these aggressive tendencies were mellowed if Prozac- a drug which boosts serotonin "uptake"- was given.

However, subsequent analysis showed significantly lower than normal serotonin levels in the neural connections of the hamster's brains. This suggests there may be a longer-term effect of taking steroids while the brain is still developing. Professor Richard Melloni, who helped run the study, was quoted as saying by BBC:
"we know testosterone or steroids affect the development of serotonin nerve cells, which, in turn, decreases serotonin availability in the brain. The serotonin neural system is developing adolescence and the use of anabolic steroids during this critical period appears to have immediate neural and behavioral consequences."

7. The thrust area of the research mentioned in the passage was _____.
a. The effect of serotonin on the human brain.
b. the effect of high doses of anabolic steroids.
c. The immediate neural & behavioral consequences of the use of anabolic steroids.
d. The development of the serotonin nerve cells.
e. all of the above

8. Why were hamsters considered for the experiments?
a. They were similar to humans in their metabolic function.
b. They were easily available
c. they have a similar neurological circuitry as Human beings.
d. The effect of steroids on them is similar as on human beings.
e. a, c & d.

9. Which of the following sentences is true, according to the passage?
a. Adolescents are more likely to have neural and behavioral disorders.
b. Depression and aggression are caused by a lower intake of serotonin.
c. Those taking steroids are likely to face long-term neural and behavioral implications.
d. Higher serotonin intake is good for brain function.
e. none of these

10. The drug that boosts serotonin uptake is _____.
a. Prozac b. Melanin c. Erythrocytes d. Penicillin
e. None of these

11. Why do adolescents develop neural disorders?
a. The effect of steroids hampers the growth of the brain.
b. Prozac, if taken in excess by adolescents, makes them aggressive.
c. Due to a decrease in the level of serotonin in the blood.
d. Steroids effect the under developed brain more.
e. None of these.

12. Please give me _____cake that is on the counter.
a. a b. an c. the d. no article needed

13. This bicycle ---- in our family for fourteen years. My father used it for the first five years, my brother rode it for the next five and I ----it for the last four.
a. is/have had b. had been/had c. has been/have had d. was/had had

14. She was mopping the floor when her cousin came in. She _____ him angrily because he made the floor dirty with his muddy shoes.
a. was scolding b. had scolded c. will scold d. scolded

15. 3C's of Précis writing are-
a. Comprehension, condensation and coverage b. Comprehension, clarity and coverage
c. Condensation, clarity and coverage d. Comprehension, condensation and clarity

16. My school _____ a food-and-fun fair next month to raise money for the school building-fund.
a. has been holding b. is holding c. held d. has held

17. Last year, I experienced how tedious long plane trips could be. I ---- in an airplane for fairly long distances before that, but never as long as when I went to Australia last June.
a. have never flown b. didn't fly c. have flown d. had flown

18. Letters A, B and C in ABC of writing denotes-
a. accuracy, brevity and clarity
b. assertiveness, brevity and clarity
c. assertiveness, brevity and creativity
d. accuracy, brevity and creativity

19. Identify the Essay type- Is Engineering still considered the best career option?
a. Persuasive Essay b. Descriptive Essay c. Comparative Essay d. Evaluative Essay

20. When superiors provide directions to their subordinates regarding what to do, this is known as a type of _____ communication.
a. upward b. downward c. horizontal d. lateral

21. It's a great pity you didn't come to London with us last summer, As you --- it before, it ---- a wonderful holiday for you.
a. hadn't seen/would have been
b. have never seen/will surely be
c. didn't see/has been
d. didn't see/had been

22. John likes to play _____volleyball.
a. a b. an c. the d. no article needed

23. The meaning of the word convention is:
a. a collection of recorded music
b. any business that produces goods or provides services; the work and related activity in factories and offices
c. a large meeting for a special purpose
d. to take and hold or to control by force

24. Mary was accompanied _____ her mother to the concert.
a. to b. by c. with d. from

25. Our neighbours have _____cat.
a. a b. an c. the d. no article needed

26. My daughter is learning to play _____violin at her school.
a. a b. an c. the d. no article needed

27. Active listening requires
a. The person to listen carefully all the time.
b. The person seeking assistance to express himself clearly.
c. Everyone to concentrate and ask questions.
d. The person to sit forward and have eye contact with the person speaking.
e. Active engagement with the person and his message with appropriate non-verbal cues.

28. In the evenings, I often play chess with my next door neighbour. I ---- chess with him ever since I ---- to live here ten years ago.
a. have been playing / came
b. play / came
c. have been playing / have come
d. am playing / have come

29. Has the taxi come yet? It is already ten minutes late. I hope it _____ here soon.
a. will be b. is c. was d. would be

30. To get to the Marketing department, you have to go _____ those stairs and then _____ the corridor to the end.
a. Up/along b. between/along c. out of/between d. out of/into

31. He drove _____ me without stopping and drove off _____ the centre of town.
a. towards/over b. along/up c. past/towardsd. in/next to

32. I couldn't get in _____ the door so I had to climb _____ a window.
a. through/in b. between/into c. out of/between d. out of/into

33. She took the key _____ her pocket and put it _____ the lock.
a. between/into b. out of/in c. by/on d. up to/out of

34. Choose the correct sequence for the stages of Listening.
1. Receiving 2. Attending 3. Understanding
4. Remembering 5. Evaluating 6. Responding
a. 123456 b. 143562 c. 324561 d. 124563

35. Why is business writing important for a professional?
I. Need to build up skills learnt in school and college.
II. It's an integral part of corporate life.
III. Will help in personal life.
IV. Will help write reports and e-mails well.
a. I, II, III b. II and IV c. III and II d. I and IV

36. Ted's flight from Amsterdam took more than 11 hours. He _____be exhausted after such a long flight. He might prefer to stay in tonight and get some rest.
a. might b. must c. have to d. should

37. Which skill is not involved in active listening?
a. Emphasizing b. Suspending Judgement c. Summarization d. Paraphrasing

38. I thought that the grass _____ cutting, but the lawnmower was out of order. Therefore, I cut the grass with a pair of shears.
a. had needed b. needs c. needed d. will need

39. A comparative essay -
a. demonstrates resemblances and variations between two topics.
b. contains detailed description.
c. evaluates a particular topic.
d. promotes one idea over another by using persuasive techniques.

40. _____ refers to words and phrases that are regarded as very informal and are more common in speech than writing.
a. Slang b. Colloquialisms c. Jargon d. Indianism

41. A descriptive essay -
a. demonstrates resemblances and variations between two topics.
b. contains detailed description.
c. evaluates a particular topic.
d. promotes one idea over another by using persuasive techniques.

42. Choose the correct spelling word.
a. appollogy b. apollogy c. appology d. apology

43. The meaning of the word march is
a. a land area with water all around it
b. to free from danger or evil
c. to go up; to go higher
d. to walk in a group like soldiers; to walk together in a large group to protest about something

44. He made his escape by jumping _____ a window and jumping _____ a waiting car.
a. over/into b. between/into c. out of/between d. out of/into

45. When sending business mails it is important to -
a. use coloured text b. use 'Urgent' as the subject
c. be clear and concise d. use Bcc

46. The meaning of the word apologize is:
a. a grain used to make bread; the plant that produces the grain
b. a part of something larger
c. the general condition of the body and mind; the condition of being free from sickness or disease
d. to express regret for a mistake or accident for which one accepts responsibility ·

47. Which of these is the first step to writing an essay?
a. Think about an answer
b. Analyse the question
c. plan the answer
d. write the answer

48. Dave: _____you hold your breath for more than a minute?
Nathan: No, I can't.
a. could
b. can
c. have to
d. should

49. I _____speak Arabic fluently when I was a child and we lived in Egypt. But after we moved back to Canada, I had very little exposure to the language and forgot almost everything I knew as a child. Now, I can just say a few things in the language.
a. could
b. can
c. should
d. have to

50. Please make sure to water my plants while I am gone. If they don't get enough water, they _____ die.
a. might
b. must
c. can
d. have to

51. When you have a small child in the house, you _____ not leave small objects lying around. Such objects can be swallowed, causing serious injury or even death.
a. can
b. must
c. have to
d. should

52. Choose the odd connecting word-
a. In addition to
b. Furthermore
c. Besides
d. On the contrary

53. I saw something about it _____ television.
a. in
b. on
c. at
d. through

54. We _____TV when it started to rain.
a. watched
b. were watching
c. are watching
d. had watched

55. Have you visited any European countries? - Yes. I _____ Spain and Italy two years ago.
a. have visited
b. had visited
c. visited
d. was visiting

56. Which of the following is not one of the 7C's of clarity?
a. Creative
b. Concise
c. Credible
d. Courteous

57. Since 2011 they _____ their son every year.
a. had visited
b. have visited
c. are visiting
d. visited

58. I want _____apple from that basket.
a. a
b. an
c. the
d. no article needed

59. KISS concept in Business Writing is -
a. Keep it Simple, straight, strong and short
b. Keep it simple, short, strong and sincere
c. Keep it short, simple, straight and sincere
d. Keep it short, simple, stylish and strong

60. _____ is an automatic psychological process of receiving oral messages.
a. Speaking
b. Listening
c. Hearing
d. None of these

61. Where is he? I _____ for him since three o'clock!
a. am waiting
b. have been waiting
c. was waiting
d. had been waiting

62. Effective listening requires skill and often practice. To listen effectively you must
a. Understand only the words.
b. Understand only the emotions.
c. Adapt to each individual and context, and engage with the person and all parts of their message.
d. Demonstrate respect and trustworthiness.
e. Develop a therapeutic relationship.

63. An individual can recall _____ of what he/she has just heard.
a. 50%
b. 45%
c. 60%
d. 55%

64. Good listeners often
a. take over the conversation
b. change the subject
c. Ask few questions
d. ask for opinions, views and feedback

65. I bought _____umbrella to go out in the rain.
a. a b. an c. the d. no article needed

66. You write a memo -
a. To persuade action b. To issue a directive c. To provide a report d. All of the above

67. An appropriate ending for a letter to your boss requesting for a leave would be-
a. Yours lovingly b. Yours Faithfully c. Regards d. Yours thankfully

68. I envy you. At five tomorrow you _____ some tan on a nice beach in Greece.
a. will get b. will be getting c. will have gotten d. will have been getting

69. Choose the correct spelling word.
a. permitted b. pearmitted c. permytted d. permited

70. The meaning of the word brief is
a. a prison for those waiting to be tried for a crime or for those serving sentences for crimes that are not serious
b. to move into or be in a position in which only the feet are on a surface; to be in one position or place
c. a device of strong, light material that rises when filled with gas lighter than air
d. short; not long

71. While writing an e-mail you should keep tour message short and concise.
a. True b. False

72. While writing a précis you
a. Condense the main idea
b. take the basic idea and elaborate ideas
c. change the overall idea to make it interesting
d. None of the above

73. Choose the correct spelling word.
a. propagate b. propogate c. propagait d. prapagate

74. The following introduction is appropriate for which of the following essay types-
Education is the basic need of a person. Regular education i.e. in educational institutions has been popular for centuries together. But in the present circumstances distance education has also become popular and the need of the time. Both regular and distance systems of imparting education have their own advantages.
a. Persuasive Essay b. Comparative Essay c. descriptive Essay d. Evaluative Essay

75. Choose the correct spelling word.
a. significanse b. segnificance c. significance d. sagnificance

76. Humor and sarcasm should be included in an e-mail as it makes it interesting.
a. True b. False

77. You have to remind your employees to attend an important meeting that no one wants to attend. You would write-
a. a memo b. a report c. an e-mail d. a letter

78. The following introduction is appropriate for which of the following essay types-
There should be no dispute as regards banning smoking at public places. There is hardly anyone who is not aware of the health hazards posed by smoking. Smoking is the main cause of heart and pulmonary diseases. It can cause nervous disorders, cancer, asthma, lack of appetite and other serious diseases. Cigarettes contain nicotine and ammonia which are very dangerous for human life.
a. Persuasive Essay b. Comparative Essay c. descriptive Essay d. Evaluative Essay

79. You must cool down before sending an e-mail to prevent flaming another user.
a. True b. False

80. While sending a non- urgent e-mail, you may keep the subject line blank.
a. True b. False

Directions for Questions (81 & 82): In the following some parts of the sentence have been jumbled up. You are required to re-arrange these parts which are labeled P, Q, R and S to produce the correct sentence.

81. Like most people,
P : all of their behaviour
Q : entrepreneurs are complex
R : can explain
S : and no one theory
The proper sequence should be:
a. QPRS b. RSQP c. QSRP d. RPQS

82. Today
P : in a fishbowl
Q : where any action may be the subject of media scrutiny
R : managers at most large organizations
S : realize that they operate
The proper sequence should be:
a. RSPQ b. QPSR c. RPSQ d. QSPR

83. Choose the correct spelling word.
a. curios b. currious c. curyous d. curious

84. The meaning of the word enzyme is:
a. to make something go or move faster
b. the group that includes children and their parents
c. acting safely; with much thought
d. a special kind of protein; it produces changes in other substances without being changed itself

85. Choose the correct spelling word.
a. absence b. absense c. abcense d. absens

86. The meaning of the word moderate is
a. a narrow way for walking; a way along which something moves
b. to offer or propose something to think about or consider
c. an examination in a court of a question or dispute to decide if a charge is true
d. not extreme

87.Choose the correct spelling word.
a. exceed b. excead c. exseed d. exeed

88. He _____out five minutes ago.
a. went b. had gone c. has gone d. will be going

89. Look! The Sun _____ over the hills.
a. rises b. is rising c. rose d. has been rising

90. Every day last week my aunt _____ a plate.
a. brakes b. broke c. was breaking d. has been breaking